Project Proposal

Agentic AI Applications Course

Team Number :

Team Member Names : Ruchika C Lal, Karthik J Ramoo, Jayashree G, Tanmay C J

Proposed Project Title : Autonomous Decision Intelligence Layer for Extraterrestrial Rovers

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1. Executive Summary

This document proposes the development of an Autonomous Decision Intelligence Layer for deep-space rovers operating in high-latency, resource-constrained, and hazardous extraterrestrial environments. The system leverages a hierarchical multi-agent agentic AI architecture to enable real-time autonomous decision-making without continuous human supervision.

The proposed system addresses critical mission challenges, including communication delays, energy optimization, terrain risk assessment, hazard detection, and scientific prioritization. By integrating structured tool usage, reflection loops, memory retrieval, and coordinated agent collaboration, the system transitions rover operations from instruction-based automation to goal-driven autonomous intelligence.

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2. Problem Statement & Use Case

Operational Constraints in Deep Space Missions

Planetary rovers operating on Mars, the Moon, and asteroids face severe constraints:

- High communication latency (4–20 minute signal delays)
- Limited bandwidth
- Finite energy reserves
- Unknown terrain risks
- Harsh environmental conditions

Traditional rover systems rely on Earth-based decision loops, significantly limiting responsiveness and scientific yield.

Current System Limitations

Existing systems depend on delayed teleoperation or static rule-based automation. These approaches:

- Slow down mission execution
- Underutilize onboard intelligence
- Prevent dynamic scientific prioritization
- Limit adaptive hazard response

Target Users

Primary users include space agencies, commercial lunar exploration firms, and planetary science research teams. Secondary dual-use applications extend to mining robotics, defense systems, underwater exploration, and disaster response robotics.

Why this requires an agentic AI solution

Rover decision-making cannot be handled by a single LLM prompt or a simple RAG chatbot because it requires iterative planning, structured tool-based numerical reasoning (terrain models, energy estimators), real-time constraint validation, and multi-domain tradeoff evaluation. A static prompt cannot dynamically replan, enforce safety constraints, or coordinate terrain, energy, hazard, and science reasoning simultaneously.

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3. Agentic Architecture Justification

This application is inherently agentic due to its need for:

- Continuous environmental observation
- Multi-step planning and decomposition
- Dynamic tradeoff evaluation
- Real-time replanning
- Tool-based structured reasoning
- Reflection and validation loops

- Long-term memory integration

Core Agentic Capabilities Required

- **Tool Use:** Integration with terrain classifiers, energy prediction models, and hazard detection systems.
- **Planning & Decomposition:** Breaking high-level mission goals into executable sub-tasks.
- **Multi-Agent Coordination:** Parallel evaluation by specialized agents with supervisor synthesis.
- **Reflection / Self-Correction:** Validation of decisions before execution.
- **Memory Integration:** Learning from past mission outcomes.

What Would Fail with a Single LLM Call?

A single LLM call would:

- Produce unverified numerical reasoning.
- Lack persistent memory of past terrain decisions.
- Fail to enforce hard safety thresholds.
- Be unable to iteratively revise plans after detecting risk.

Expected Agentic Behavior Example

For example, if the Terrain Agent reports moderate slope but the Hazard Agent detects surface instability, the system should reflect, reassess risk thresholds, and revise the route before execution rather than blindly proceeding.

4. High-Level Solution Overview

Intended Agentic Architecture

Our system is designed as a multi-agent decision-making architecture for an autonomous rover. Instead of relying on a single LLM call, we divide responsibilities across specialized agents coordinated by a Supervisor.

Main Agents and Roles

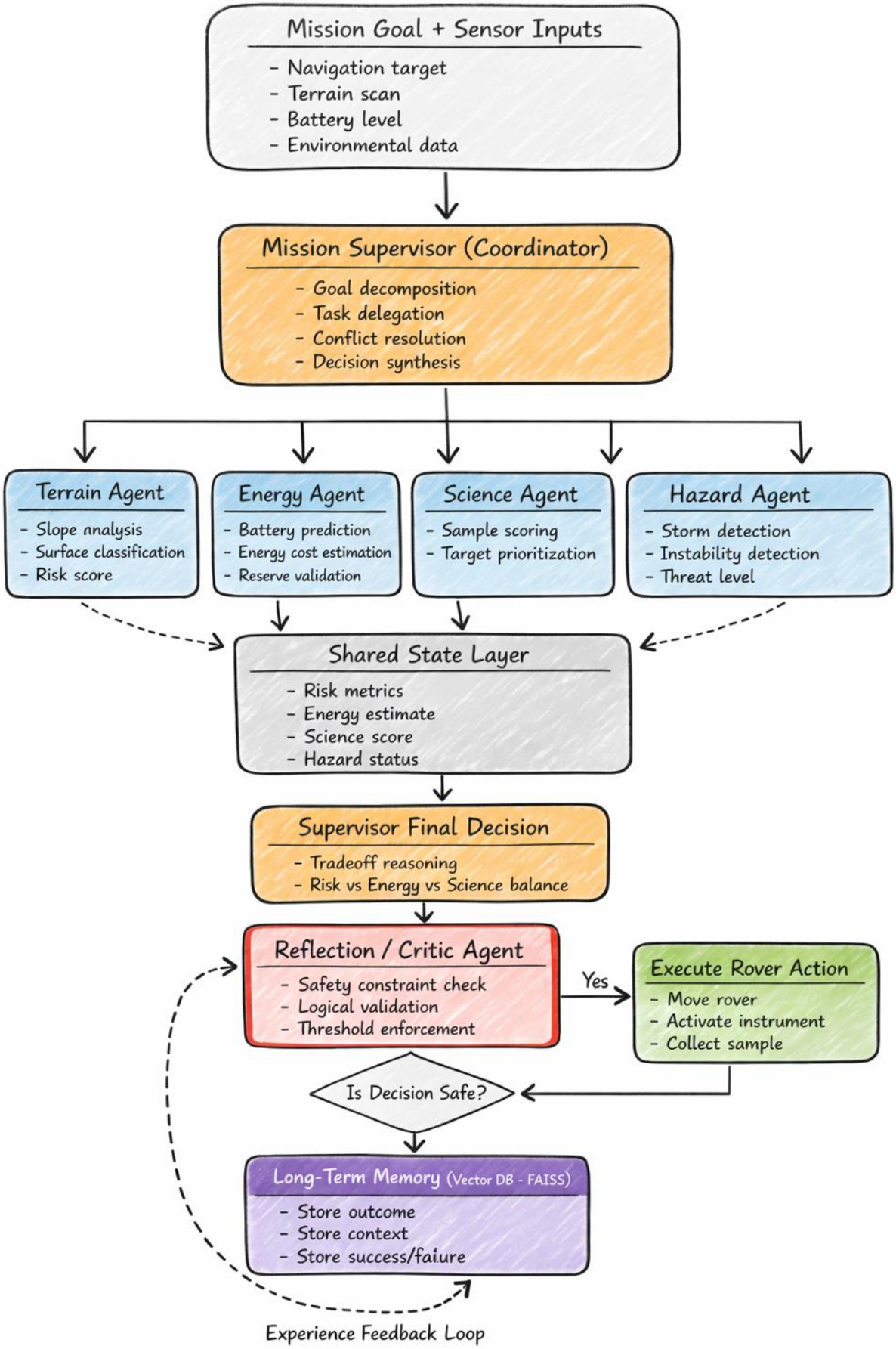
- **Mission Supervisor (Coordinator)**: Decomposes the mission goal, delegates tasks, synthesizes results, and proposes a final action.
- **Terrain Agent (Researcher)**: Evaluates slope, surface type, and produces a risk score.
- **Energy Agent (Resource Manager)**: Estimates battery usage and validates minimum energy reserves.
- **Science Agent (Evaluator)**: Scores and prioritizes scientific targets.
- **Hazard Agent (Safety Monitor)**: Detects environmental threats and assigns threat levels.
- **Reflection / Critic Agent**: Validates the proposed decision against safety constraints and logical consistency before execution.

Decision Flow

1. Environmental input is received.
2. Supervisor decomposes goals.
3. Specialized agents evaluate in parallel.
4. Shared state updates memory.
5. Supervisor synthesizes recommendations.
6. Reflection agent validates decision.
7. Approved action is executed.
8. Outcomes are logged for learning.

Key Innovations

- Goal-oriented decision intelligence
- Multi-domain reasoning
- Structured tool integration
- Reflection-based safety assurance
- Persistent episodic memory


Mission Goal + Sensor Inputs
- Navigation target
- Terrain scan
- Battery level
- Environmental data
Mission Supervisor (Coordinator)
- Goal decomposition
- Task delegation
- Conflict resolution
- Decision synthesis
Terrain Agent
- Slope analysis
- Surface classification
- Risk score
Energy Agent
- Battery prediction
- Energy cost estimation
- Reserve validation
Science Agent
- Sample scoring
- Target prioritization
Hazard Agent
- Storm detection
- Instability detection
- Threat level
Shared State Layer
- Risk metrics
- Energy estimate
- Science score
- Hazard status
Supervisor Final Decision
- Tradeoff reasoning
- Risk vs Energy vs Science balance
Reflection / Critic Agent
- Safety constraint check
- Logical validation
- Threshold enforcement
Yes
Execute Rover Action
- Move rover
- Activate instrument
- Collect sample
Is Decision Safe?
Long-Term Memory (Vector DB - FAISS)
- Store outcome
- Store context
- Store success/faiure
Experience Feedback Loop

5. Technology Stack

Core Models

- Quantized Mistral 7B (Primary LLM)
- Terrain Classification CNN / Vision Transformer
- Energy Prediction Regression Model
- Hazard Detection Classifier
- Embedding Model for Memory Representation

Frameworks & Infrastructure

- LangGraph for agent orchestration
- ROS2 for robotics middleware
- Gazebo simulation environment
- FAISS for vector memory retrieval
- PostgreSQL & InfluxDB for structured data storage

Optimization Strategy

- 4-bit quantization
- Tool-based structured outputs
- Edge deployment simulation
- Controlled iteration limits

LLM Provider & Rationale

- **Primary Model:** Quantized Mistral 7B
- Chosen for open-weight flexibility, edge deployment feasibility, and strong reasoning performance.

Frontend / Delivery

- **Streamlit Dashboard** for simulation visualization and decision trace monitoring.
- Optional CLI interface for testing.

Evaluation Tools

- Manual evaluation rubric for reasoning quality.
- Custom metrics dashboard for task success rate and latency.
- Logging system for tool-call monitoring.

Deployment Target

- Local edge simulation environment.
- Optional deployment on Hugging Face Spaces for demonstration.

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6. Evaluation & Success Metrics

Quantitative Metrics

- Task Success Rate ≥ 85%
- Hazard Avoidance Accuracy ≥ 95%
- Energy Efficiency Improvement ≥ 20%
- Decision Latency < 30 seconds
- Safety Compliance = 100%

Test Scenarios

- Terrain difficulty progression
- Energy constraint variation
- Scientific priority tradeoffs
- Emergency hazard handling
- Long-duration mission simulation

Baseline Comparisons

- Teleoperation simulation
- Rule-based automation
- Single-agent LLM decision system

Test Coverage Plan

- Minimum **20–25 structured test scenarios** by final demo.
- Include varied terrain difficulty, energy constraints, and hazard conditions.

Additional Metrics

- Tool-call accuracy (% valid structured outputs).
- Iteration count per decision (to detect loops).
- Cost estimation (token usage during simulation).

Qualitative Evaluation

- Manual rubric scoring (accuracy, safety compliance, coherence).
- Peer evaluation of 5–10 simulated mission runs.

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7. Risks & Mitigation

Key identified risks include:

- LLM hallucination
- Infinite reasoning loops
- Over-ambitious scope
- Simulation-reality gap

Mitigation strategies include structured validation, reflection agents, hard safety constraints, iteration limits, and conservative fallback mechanisms.

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8. Timeline & Milestones

By Mar 25 (Demo 1):

- Single-agent prototype with terrain tool integration.
- Basic reflection loop implemented.

By Apr 10 (Document 2):

- Multi-agent architecture defined.
- Early evaluation metrics collected.
- Updated documentation.

By Apr 22 (Demo 2):

- Multi-agent MVP operational.
- Evaluation dashboard working.
- Simulation scenarios implemented.

By May 5 (Final Submission):

- Fully evaluated system (≥20 test cases).
- Performance benchmarks documented.
- Final presentation + demo ready.

Conclusion

This project demonstrates how a multi-agent agentic AI system can enable autonomous, safety-aware, and goal-driven decision-making for extraterrestrial rovers. By integrating structured tools, reflection loops, and memory-based reasoning, the proposed system moves beyond static automation toward adaptive mission intelligence.

The architecture is technically feasible within the defined timeline and holds significant academic, strategic, and commercial potential.